MARVELL | Microsoft | ORACLE

Spearheading a New Security Standard for Data Centers

A new Open Compute Project (OCP) working group will standardize hardware-based, attested secure integrated HSM for cloud and hyperscale data center infrastructure

Technical Session: Thursday (10/15), 8:20am-8:40am | SJCC – Concourse Level-212

Forward-looking statements

Except for statements of historical fact, this presentation contains forward-looking statements (within the meaning of the federal securities laws) including statements related to future revenue, future earnings, and the success of our product releases that involve risks and uncertainties. Words such as “anticipates,” “expects,” “intends,” “plans,” “projects,” “believes,” “seeks,” “estimates,” “can,” “may,” “will,” “would” and similar expressions identify such forward-looking statements. These statements are not guarantees of results and should not be considered as an indication of future activity or future performance. Actual events or results may differ materially from those described in this presentation due to a number of risks and uncertainties.

Forward-looking statements are only predictions and are subject to risks, uncertainties and assumptions that are difficult to predict, including those described in the “Risk Factors” section of our Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q and other documents filed by us from time to time with the SEC. Forward-looking statements speak only as of the date they are made. You are cautioned not to put undue reliance on forward-looking statements, and no person assumes any obligation to update or revise any such forward-looking statements, whether as a result of new information, future events or otherwise.

© 2026 Marvell. All rights reserved.

Presenters

Marvell, Microsoft, Oracle Introductions



Cary Ussery

SVP and GM of Multi-Market Business Group at Marvell. As SVP and GM of the Multimarket Business Group at Marvell, Ussery is responsible for driving the development of the company's solutions, platforms and software offerings, leading a global engineering organization of engineering organization of over 700 employees. Previously he has held engineering leadership roles at Cavium, Celestial Semiconductor, Vivace Semiconductor, Cadence Design Systems and Intermetrics.



Bryan Kelly

Hardware Security Architect, Microsoft Azure. Driving the design and implementation of secure hardware platforms that underpin Microsoft's cloud infrastructure. With a strong foundation in hardware design and cybersecurity, Bryan combined deep technical expertise with strategic vision to ensure security is embedded throughout the hardware lifecycle—from concept through end-of-life.



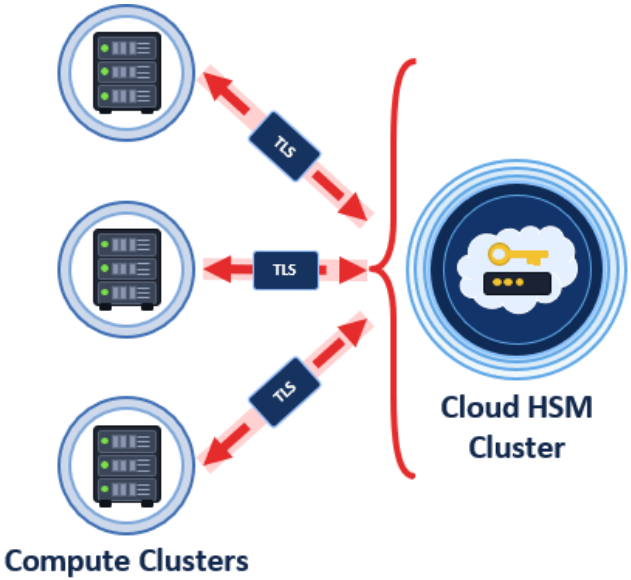
Mohan J. Kumar

Architect, Oracle Cloud. Architect at Oracle in the office of Chief Technologist. Prior to Oracle, Mohan was an Intel Fellow in Intel's Datacenter group where he led Cloud Architecture. He has close to three decades of experience in servers and has been part of many key standard such as PCIe, ACPI, and Redfish, and has been an active member and presenter at OCP! He holds close to 300 patents and is an author of a book titled "A Vision for Platform Autonomy."

© 2026 Marvell. All rights reserved.

The problem

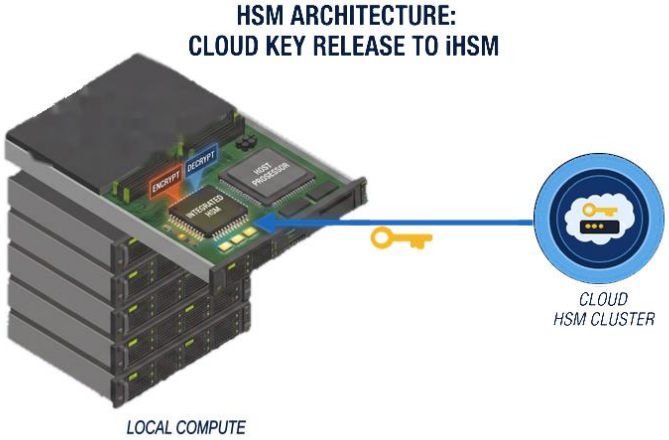
- AI and confidential compute workloads are driving the demand for cryptography at scale
- A fleet of centralized HSMs (Hardware Security Modules) alone can no longer meet this performance demand
  - Every cryptographic operation requires a network round trip to a remote HSM cluster, increasing latency
  - **Current solutions**
    1. Deal with the latency
    2. Reduce security posture: Manage the keys and crypto in software on local CPU and store the keys in unprotected memory



© 2026 Marvell. All rights reserved.

Proposed solution

- Extend the NIST FIPs 140-3 L3 boundary to the compute nodes
- A NIST FIPs 140-3 L3 integrated HSM silicon on every compute node
- No need to traverse the network, reducing latency
- Maintain security posture: Offloads the crypto from local CPU/GPU and stores keys in a protected FIPS L3 environment



© 2026 Marvell. All rights reserved.

Proposed initiative

Today’s announcement



New OCP working group

Marvell, Microsoft and Oracle form a founding working group together under OCP’s Security Track to standardize a silicon-based, NIST FIPs 140-3 L3 certified, attested, secure Integrated HSM for cloud and hyperscale data center



Strengthening cloud security

Hardware based, secure key caching, and attestation-based key lifecycle management at the server level to reduce latency, improve scalability, and strengthen the security posture of cloud workloads



Open technical specification

An initial draft specification is released—opening the door to broad industry adoption of in-server security for data centers

© 2026 Marvell. All rights reserved.

Tessera: An open Integrated HSM standard

Tessera is the open OCP standard that defines the integrated HSM



FIPS 140-3 Level 3

A dedicated security module co-located with the workload in the server tray



Open-Silicon Model

The same open model Caliptra brought to hardware root of trust, now for keys



Extends Central HSM

Extends the centralized HSM trust boundary, so keys are maintained and used locally on the tray, next to the workload

Unveiled today at OCP Global Summit 2026—jointly founded by Marvell, Microsoft and Oracle

© 2026 Marvell. All rights reserved.

Standard alignment

Tessera aligns with established open and industry standards

Caliptra
Open-source hardware root of trust

OCP LOCK
NVMe key-management block

DC-SCM
Secure management controller & platform root of trust

SPDM / TDISP
Attestation and secure device interfaces

FIPS 140-3 Level 3
NIST cryptographic module compliance

Post-quantum ready
Hybrid (ML-KEM/ECDHE) and (ML-DSA/ECDSA)

© 2026 Marvell. All rights reserved.

What’s next | Timeline

Timeline: October

October 12-15, 2026

OCP Global Summit

San Jose Convention Center, San Jose, CA

October 2026

Initial v.05 draft spec

Released at OCP Global Summit

Following

v.1.0 spec

Joint technical specification, published with the OCP community

Join the working group through the OCP Security Project: opencompute.org/projects/security
Demos: Microsoft booth (#C73), Marvell booth (#B31)

© 2026 Marvell. All rights reserved.